Communication and Internet Technology:
Mobile IP with IPv6

Mobile IPv6 is a protocol developed as a sub-set of Internet Protocol version 6 (IPv6) to support mobile connections. It allows mobile devices to move from one network to another and still maintain existing connections.

Mobile Internet Protocol (IP) provides mobility support on the existing IP infrastructure. It is needed without any modifications for existing routers, applications or fixed end-hosts. How can then mobile IP change with IPv6?

Mobile IPv6 allows an IPv6 node to be mobile like an IPv4 node. Mobile IPv6, unlike mobile IPv4, supports additional support for mobility, route optimisation and security. IPv6 allows Internet nodes to associate a mobile node's (MN's) home address with its care of address (COA) and send packets directly to COA (and not via home address).

MN obtains COA, eliminating the need for foreign agents. Mobile IPv6 uses route optimisation by which MN sends its current COA to the correspondent node. When it knows about MN's current COA, a correspondent node can deliver packets directly to MN's home address without any assistance from home agent.

Since mobile IPv6 does not make use of foreign agents, it uses correspondent node and binding. Binding refers to an association of home address with COA of MN.

Correspondent node is a peer node with which MN communicates. It may either be mobile or stationary. Its functional operation is given below.

- MN gets registration to foreign network (FN) and gets COA as in mobile IPv4.